Ending Affirmative Action Harms Diversity Without Improving Academic Merit

Jinsook Lee, Emma Harvey, Joyce Zhou, Nikhil Garg, Thorsten Joachims, and René F. Kizilcec. ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO 2024).

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What happens when race is removed from an existing ML ranking algorithm?

Does going race-unaware make admissions more ‘merit-based’—or just less diverse?

Background: Admitting applicants with academic merit.

- Colleges review tens of thousands of applications in order to fill just a few thousand slots.
- The suspension of standardized tests like the SATs during the COVID-19 pandemic led many colleges to go **‘test optional.’**
- **Common App:** a platform through which millions of college applications in the U.S. are submitted annually.
- **Applicants are embracing these test-optional policies**: On Common App 76% of applicants submitted test scores in 2019, compared with just 45% in 2023.
- As a result, **colleges no longer have reliable access to a key piece of nationally standardized information**.

Background: Admitting a diverse class

SUPREME COURT

June 2023

Supreme Court strikes down college affirmative action programs

The ruling is the culmination of decades of effort to end the consideration of race in admissions.

Students for Fair Admissions SFFA policy change

Research Questions

RQ1: How does a change in admission policy impact a college's overall class?

RQ2: How does a change in admission policy impact individual applicants?

Cannot observe actual admissions outcomes under different policies.

As a proxy, rely on **applicant ranking algorithms**, which can be used to determine the order in which applicants are reviewed.

Background: Machine learning in admissions processes.

GRADE: a tool used for graduate admissions at universities,

- reviewers validate applicants to whom GRADE had given very high or low scores and focus most of their efforts on reviewing applicants about whom GRADE was unsure.

- GRADE was abandoned in 2020 due to the concerns that it was reinforcing historical biases in admissions.

Data and Methods

- Highly selective, engineering-focused US university.
- 4 years of Regular Decision applicants: 59,833 applications (2019–20 to 2022–23).

- Common App features of first-year applicants’
    - Academics: grades, coursework, tests
    - Extracurricular activities
    - family and demographic background (incl. race/ethnicity, citizenship, and parental education), essays, and letters of recommendation.

https://www.commonapp.org/

Data and Methods: Simulating Policy Changes

- Included all data that was common across all years, except for personally identifiable information (name, date of birth, contact information, etc.).
- Train on first 3 years (2019-2022), evaluate on 2022–23 cohort.

Sample	# applicants	% accepted	% waitlist	% URM	% female	% FG	% LI
Train	44,293	5.7	14.4	17.3	31.1	15.9	25.7
Test	15,540	5.0	16.1	16.3	32.1	19.5	32.3

Baseline and policy variants

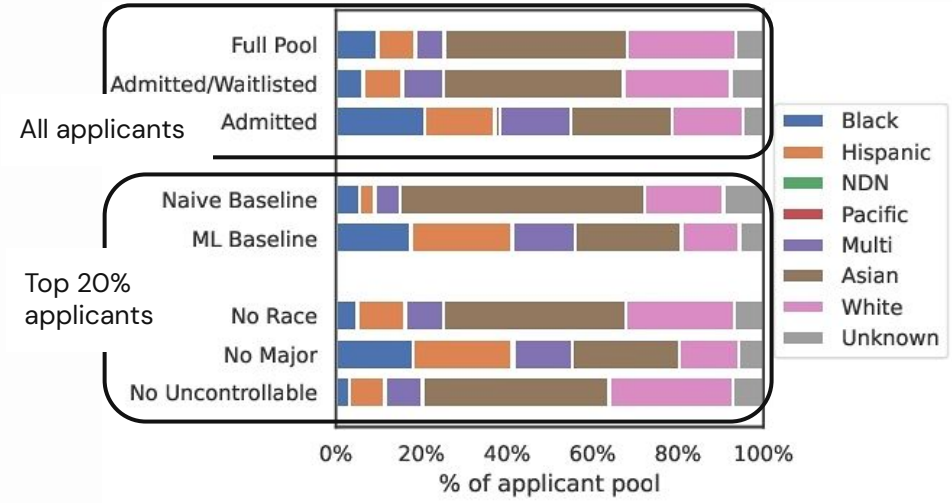
- ML baseline (race-aware): uses Gradient-Boosted Decision Tree.
  - Uses all processed features, including race/ethnicity and URM (under-represented minorities) status.
- Naive baseline: ranking using prior math level and standardized test scores.

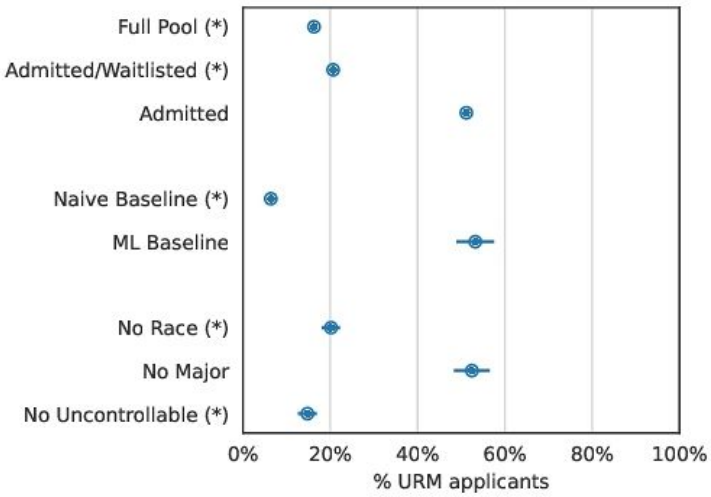
- Applicants sorted into 10 deciles by predicted probability.
- Top pool = deciles 9 & 10 → top 20% to be reviewed first by admissions officers.

Policy simulations:

- No race: drop race, ethnicity, URM features (12 features, SFFA-compliant).
- No major: drop intended major (non-protected).
- No uncontrollable (29 features): drop race, gender, citizenship, parental education, school type, socioeconomic status.

Group Impact: Academic Merit and Diversity



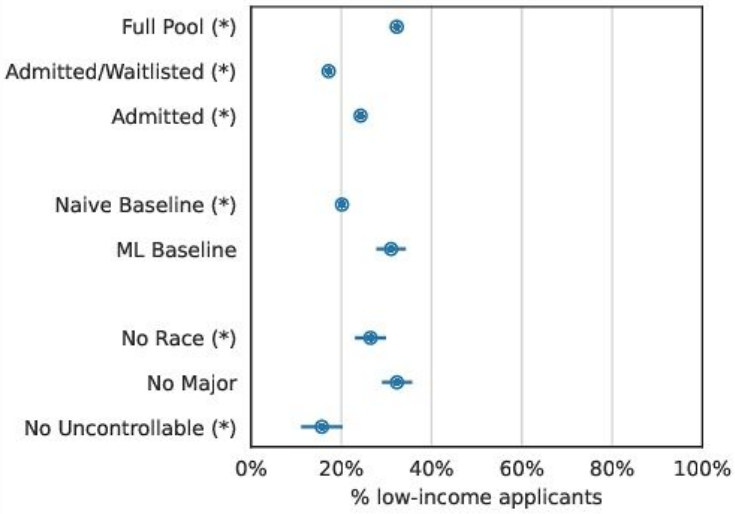


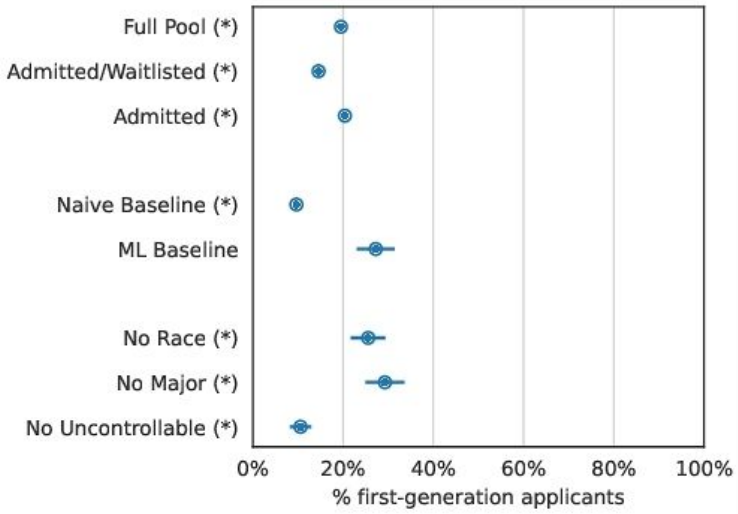
under-represented minorities

- Full pool ≈ 16% URM; admitted pool ≈ 51% URM.
- ML baseline top pool: 53% URM → matches admitted pool.
- No race: URM share in top pool drops to 20% (≈62% reduction).
- No uncontrollable features: URM share ~15%.
- No major: URM share ~52% (no meaningful change).

Compliance with the SFFA policy change significantly reduces the diversity of top-ranked applicants.

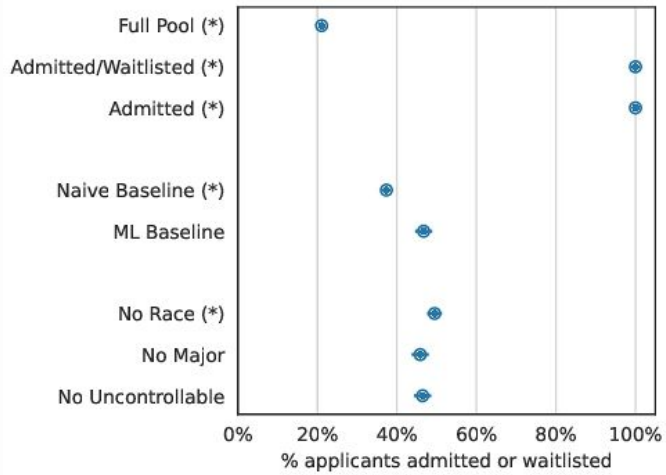
Group Impact: Academic Merit and Diversity

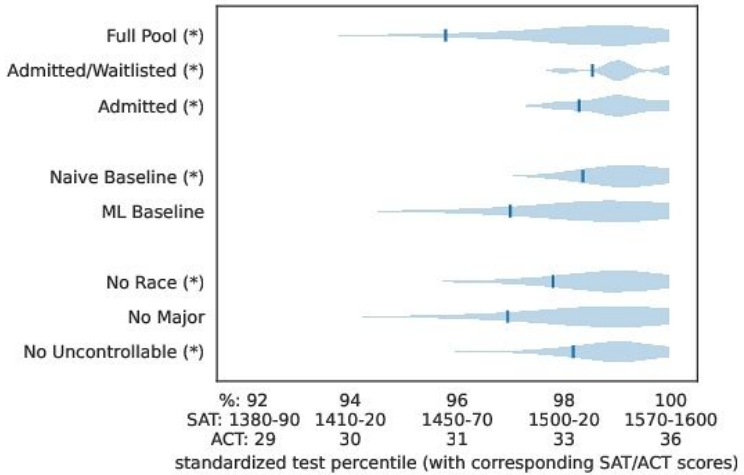




- ML baseline top pool: LI ≈ 31%, FG ≈ 27%
- No race: LI → 26%, FG → 26%.
- No uncontrollable: LI → 16%, FG → 11%.
- No major: LI ≈ 32%, FG ≈ 29% (small changes).

Group Impact: Academic Merit and Diversity





The reduction in diversity is not associated with a corresponding increase in academic merit of top-ranked applicants.

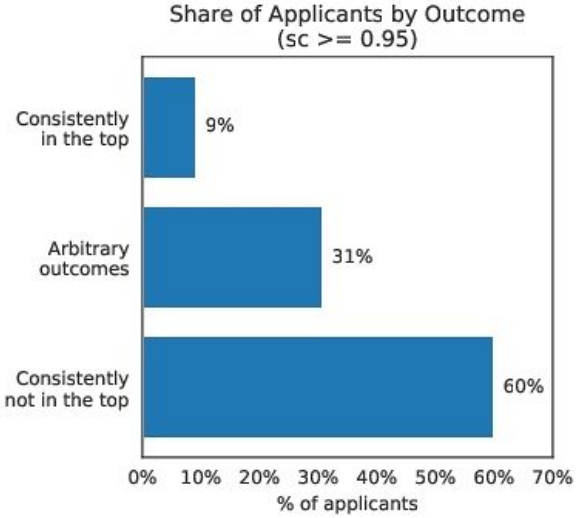
Share of top pool that are actually admitted or waitlisted:

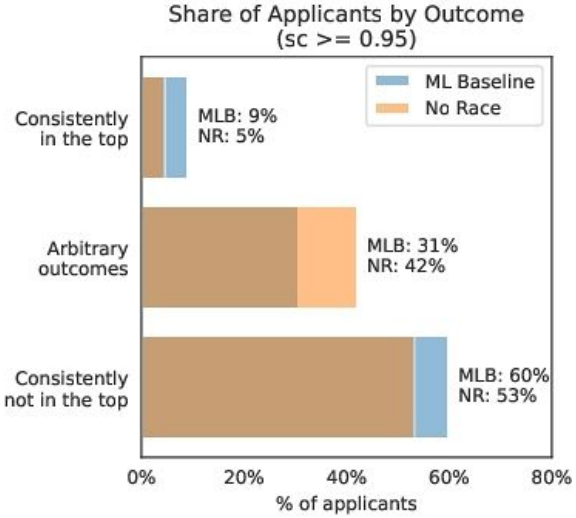
- ML baseline ≈ 47%
- No race ≈ 49% (statistically >, but marginal)
- No major ≈ 46%; no uncontrollable ≈ 46%; naive baseline ≈ 37%

Standardized test percentiles (among applicants who submitted standardized test scores):

- Admitted applicants ≈ 98.3 percentile
- ML baseline top pool ≈ 97.0; no race ≈ 97.8. Abs difference ≈ 1480 vs 1490 SAT-equivalent (tiny in practice.)

Individual Impact: arbitrariness





For each policy, train 1000 bootstrap models on slightly different samples.

- For each applicant, compute self-consistency: fraction of models that agree on whether they're in the top 20%. High self-consistency → robust outcome; low self-consistency → arbitrary.
- more than half of the top pool consists of applicants who have been added to that pool somewhat arbitrarily.
- Compared to the ML baseline model, the 'no race' model has lower self-consistency.

Question: What are the limitations of this research work?

means,

what else could have been added, what could have been done differently? Imagine if you have to do a similar study, what changes will you make in this research plan?